# ESAM-CD: Fine-Tuned EfficientSAM Network With LoRA for Weakly Supervised Remote Sensing Image Change Detection

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***Abstract*— Change detection (CD) has become an attractive research topic in the field of remote sensing imagery in recent years. Despite significant advancements driven by deep learning (DL) techniques, most current methods predominantly rely on fully supervised strategies. These methods require the collection of a large number of pixel-level labels, which is quite time consuming and label intensive. To address that, we propose a weakly supervised CD method with EfficientSAM (ESAM)-CD termed, which leverages multiscale class activation map (CAM) fusion and a fine-tuned EfficientSAM's image encoder. First, we construct a classification model employing image-level labels with a deep supervision strategy to generate high-quality multiscale CAM. Subsequently, a multiscale CAM fusion module is proposed to refine the boundaries of change targets by harnessing information from various scales. Then, we utilize EfficientSAM with powerful generalization capabilities as the backbone and fine-tune it using a low-rank adaptation (LoRA) strategy to establish a CD network. In such a network, bitemporal images and the generated pseudolabels are fed into the network. In addition, to overcome the reliance of EfficientSAM's decoder on prompts, we propose a prompt-free decoder based on the general convolutional layers to predict change maps. Finally, we validate the effectiveness of the proposed ESAM-CD using two publicly available CD datasets (i.e., WHU-CD and LEVIR-CD). Comprehensive experiments demonstrate that our method outperforms other weakly supervised CD methods, achieving outstanding performance on both datasets.**

***Index Terms*— Change detection (CD), EfficientSAM, low-rank adaptation (LoRA), multiscale class activation map (CAM), weakly supervised.**

## I. Introduction

CHANGE detection (CD) is the process of identifying the land surface difference between bitemporal remote sensing images over the same geographical region [1]. Remote sensing images span a range from low and medium resolutions to high resolution, thanks to the continuous advancement of sensor imaging techniques. Consequently, CD has further garnered the attention of numerous researchers due to the exceptional wealth of remote sensing images, and it is widely employed in various real-world applications, including land use management [2], disaster assessment [3], and urban change analysis [4].

Over the past several decades, researchers have conducted numerous studies and made significant advancements in the field of CD. Depending on the need for manual feature extraction, current CD algorithms can be categorized into two types: traditional methods and deep learning (DL)-based methods [5]. Although traditional methods have delivered remarkable performance with low- and medium-resolution remote sensing images, their effectiveness tends to be suboptimal when applied to high-resolution remote sensing images [6].

In recent years, data-driven DL techniques have advanced rapidly, propelled by the impetus of big data and artificial intelligence technology [7]. Consequently, these techniques have extended from the field of computer vision to the remote sensing community, leveraging the remarkable feature extraction capabilities of DL-based neural networks [8], [9]. Specifically, DL-based methods offer the advantage of automatically extracting multilevel abstract features, which are particularly robust for complex ground objects in remote sensing images. Moreover, DL-based CD methods benefit from an end-to-end architecture, reducing the need for manual intervention and mitigating the accumulation of errors [10].

In the early stages of research, CD is conceptualized as a semantic segmentation task. Researchers preprocessed bitemporal images using fusion strategies, such as stacking and differencing before inputting them into a segmentation network for analysis [11]. Subsequently, Daudt et al. [12] introduced the Siamese network architecture to CD, where bitemporal images are encoded by two identical branches within the designed network. This network has since become a benchmark for CD tasks, inspiring numerous researchers to propose CD methods based on the Siamese network [13], [14], [15]. Even though DL-based CD methods have achieved great success, most current CD methods predominantly rely on fully supervised strategies. In other words, these methods require collecting a large number of pixel-level labels, which is quite time-consuming and label-intensive in practice.

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To overcome the challenge of extensive pixel-level labels, researchers have turned their focus toward weakly supervised methods. In particular, the weakly supervised methods employ incomplete or imprecise annotations, which are usually in a "weaker" form and include bounding boxes, scribbles, and image-level labels. Image-level labels only use a binary indicator, typically "1" or "0," to denote the presence or absence of change between bitemporal images, and they have become the most popular weak label due to their ease of acquisition. For example, Kalita et al. [16] proposed a weakly supervised CD network that utilizes image-level semantic labels to train the network for generating the difference image (DI). In addition, inspired by the localization capability of the class activation map (CAM) [17], researchers have introduced CAM into the field of CD. For example, Jiang and Tang [18] proposed a weakly supervised approach that employs image-level labels to produce localization maps from CAM. Similarly, Wang et al. [19] proposed CS-WSCDNet, a weakly supervised CD method that leverages image-level labels and CAM. This method incorporates a classifier leveraging image-level labels to generate CAM, and then, a reactivation strategy is proposed to make CAM more attentive to changes. While these methods demonstrate progress in weakly supervised CD, a common issue among them is that they all rely exclusively on the final convolutional layer of the classifier to generate CAM. As a result, the generated CAM can only identify coarse regions of change, often missing fine details due to their low spatial resolution, which in turn affects the overall accuracy of the CD.

Recently, the emergence of foundation models such as the segment anything model (SAM) [20] and its variants (e.g., FastSAM [21] and EfficientSAM [22]) has gained great research focus in the field of computer vision. In contrast to ordinary neural networks, SAM is trained on the world's largest segmented dataset, and its remarkable segment capability in natural image scenarios has been well-validated across a variety of scenarios [20]. Therefore, researchers have introduced SAM and its variants into the field of CD due to their powerful visual recognition and segmentation capabilities. For example, Ding et al. [23] proposed a CD method based on FastSAM, capitalizing on its strong visual recognition capabilities. Furthermore, this method introduces a convolutional adaptor designed to enable FastSAM to concentrate on changed objects within remote sensing images. Mei et al. [24] proposed a semantic CD method by combining MobileSAM and convolutional neural networks (CNNs). This method leverages MobileSAM and CNNs to extract global semantic change features and local detail change information, respectively. After that, a semantic adaptor is proposed to aggregate the semantic-oriented information about changing objects. Sun et al. [25] proposed a CD method based on SAM. In this method, rich semantic knowledge is mined from SAM and then used as constraints for interested change categories.

While significant progress has been made, some challenges remain in applying SAM to the field of CD. On one hand, there exists a significant imaging difference between natural images and remote sensing images, which hinders the satisfactory performance of SAM when directly using pretrained weights for CD. However, retraining all the parameters in the SAM on personal devices is challenging due to the huge number of parameters involved. On the other hand, SAM is a segment model that employs prompts (e.g., points, bounding boxes, and masks) to generate target segmentation results [20]. Consequently, it is difficult to provide a suitable prompt when it is uncertain whether change regions exist in the bitemporal images and where these change regions are located.

Motivated by the aforementioned challenges, this work proposes a novel weakly supervised CD method named EfficientSAM (ESAM)-CD. First, a classification network is constructed using image-level labels to generate CAM. To improve the quality of the generated CAM, we introduce the deep supervision strategy for multiscale CAM and propose a multiscale CAM fusion strategy that leverages information from different scales of CAM to enhance the boundary details of change regions. Subsequently, a CD network is constructed based on EfficientSAM. To tailor its feature extraction capability for remote sensing images, we integrate the low-rank adaptation (LoRA) technique to fine-tune the image encoder of EfficientSAM. Furthermore, to address the reliance of EfficientSAM's decoder on prompts, we propose an innovative prompt-free decoder built upon general convolutional layers. Consequently, we explore the potential of integrating this decoder based on general convolutional layers with the fine-tuned EfficientSAM's image encoder for remote sensing image CD tasks. Our contributions can be summarized as follows.

1) We propose a novel weakly supervised CD method named ESAM-CD based on a fine-tuned EfficientSAM using LoRA. This method can overcome the limitation that the fully supervised method requires extensive pixel-level labels by leveraging only image-level labels during training.
2) To address the loss of boundary details in CAM generated from the deepest convolutional layer, we introduce the deep supervision strategy to generate high-quality multiscale CAM. Subsequently, a multiscale CAM fusion strategy is proposed to capture accurate boundary details for generating pixel-level pseudolabels.
3) To improve the feature extraction capabilities of EfficientSAM's image encoder for remote sensing images, we freeze the image encoder and introduce the LoRA technique to fine-tune it. Furthermore, we propose a novel prompt-free decoder based on general convolutional layers to explore the potential integration of general convolution with EfficientSAM for CD.

The remainder of this article is structured as follows. Section II presents a brief review of related works in CD. Section III details the proposed method, ESAM-CD. Section IV outlines the experimental procedures and discusses the results. Finally, Section V concludes this article.

## II. Related Works

Over the past few years, numerous CD methods based on DL have emerged. Depending on whether pixel-level label information is provided to the network, these methods can be divided into four groups: fully supervised, unsupervised, semi-supervised, and weakly supervised.

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### A. Fully Supervised CD Methods

A significant portion of current research on CD methods concentrates on fully supervised methods, particularly those utilizing an encoder–decoder architecture to train networks for identifying changed regions. In the early stages, researchers primarily employed CNNs and proposed various CD methods. For instance, Daudt et al. [12] introduced a shared-weights dual-branches network based on the Siamese U-Net architecture. This design has since become a benchmark for CD methods due to the Siamese network's unique dual-branch structure, which is particularly well-suited for processing bitemporal images. Liang et al. [26] proposed a CD method that integrates the Siamese network with metric learning. This method extracts pyramid features from the bottleneck layer of the Siamese network, after which a hierarchical metric learning strategy is employed to learn more discriminative and representative features. Yu et al. [27] developed a mask classification-based CD (MaskCD), approaching the problem from a mask prediction and classification perspective. Specifically, this method aims to identify change regions from bitemporal images at the mask level by adaptively predicting and classifying the masks to change regions.

To overcome the inherent drawback of the local receptive field of CNN, which limits their ability to capture features related to the global context [28], some researchers have introduced transformers that excel at modeling spatial–temporal relationships and capturing long-range features using a self-attention mechanism [29]. For example, Zhang et al. [30] proposed a CD method with a Siamese structure that utilizes the Swin Transformer blocks as the basic units to extract multiscale features. To leverage the advantages offered by both transformers and CNNs, several studies have integrated these architectures for CD tasks. For instance, Chen et al. [31] proposed a full-scale connected CNN-transformer network for CD. This approach incorporates a progressive attention module to fuse features extracted from both CNNs and transformers. Yin et al. [32] developed a hybrid CNN-transformer method that employs a tokenizer to embed CNN-extracted features into a sequence of tokens. The transformer module in this approach is then utilized to model global spatiotemporal context within the token space. In conclusion, although fully supervised methods have achieved significant progress, the key challenge lies in the fact that these methods require highly accurate pixel-level labels.

### B. Unsupervised CD Methods

Unsupervised CD methods do not require any labeled information or prior knowledge [33]. In the early stages of this article, the majority of unsupervised methods are based on threshold segmentation and clustering algorithms [34]. Specifically, these methods consist of two key stages. First, a DI is generated by comparing bitemporal images. Subsequently, a change map is derived through the analysis of the DI using threshold segmentation and clustering algorithms. However, these methods are developed based on hand-designed features, which are highly susceptible to the influence of imaging factors.

With the advancement of DL techniques, unsupervised CD methods have incorporated DL into traditional approaches. For example, Gong et al. [35] introduced generative adversarial networks (GANs) into CD and proposed an unsupervised method based on GANs. In this method, CD is approached as a generative learning procedure, employing GANs to model the relationship between bitemporal images and the desired change map. Saha et al. [36] integrated deep features extracted from a CNN model with change vector analysis to propose an unsupervised method known as deep change vector analysis. Wu et al. [37] introduced a kernel principal component analysis convolution network for unsupervised CD. This method utilizes a deep Siamese network to extract high-level spatial–spectral features, followed by threshold segmentation to generate change maps. While these studies have been proposed for unsupervised CD methods, the accuracy of these methods still exhibits substantial room for improvement due to the absence of labeled information in the models.

### C. Semi-Supervised CD Methods

Semi-supervised CD methods lie at the intersection of supervised and unsupervised learning, leveraging a limited amount of labeled information with a large amount of unlabeled information to train CD models [38]. Given their capacity to mitigate the reliance on extensive labeled information, semi-supervised methods have garnered significant attention from researchers. For example, Zhang et al. [39] proposed a coarse-to-fine method that uses the coarse detection result generated from the distance metrics and neighborhood information to guide the final detection process. Wang et al. [40] introduced a semi-supervised method based on reliable contrastive learning. Specifically, this approach proposes a contrastive loss to identify the change targets for feature extraction and subsequently selects reliable data from unlabeled samples to generate pseudolabels for model training. Peng et al. [41] introduced a semi-supervised method based on a fully convolutional network and a GAN. To improve detection results, this approach employs two discriminators to improve the distribution consistency of segmentation adversarial loss and entropy adversarial loss between labeled and unlabeled samples. In contrast to previous approaches, Wang et al. [42] developed a general and convenient semi-supervised CD framework based on a dual shared network and dual branching networks. This framework integrates both pixel-level and patch-level annotations, progressively enhancing the consistency of change feature distribution between labeled and unlabeled data during training. While semi-supervised CD methods have yielded promising results, further investigation is warranted regarding two key aspects: the potentially destabilizing effect of unlabeled samples on the model training process and the intrinsic complexity of semi-supervised models.

### D. Weakly Supervised CD Methods

Fully supervised methods demand highly accurate pixel-level labels, while unsupervised methods often exhibit poor

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performance, constraining the development and practical application of both methods. Therefore, some researchers have shifted their focus toward weakly supervised CD, which does not necessitate massively accurate pixel-level labels. One important strategy involves first generating pseudolabels using a CAM, which is derived from natural images, to identify the target's location. For example, Jiang and Tang [18] proposed a weakly supervised method by incorporating a convolutional layer for processing the last feature maps. Subsequent to this, a global average pooling (GAP) layer is employed to generate the initial localization map. Similarly, Wang et al. [19] proposed a novel weakly supervised CD framework by combing the CAM and SAM. However, in this work, SAM is solely employed as a refined module. While the above works introduce CAM to CD, they solely rely on the last feature maps, which have limited capability to capture rich edge information about target regions. To address this challenge, researchers have proposed a series of improved CAM algorithms. For example, Zhang et al. [43] introduced a structure-aware object locating module to capture more complete target regions of each class by exploring structure similarity analysis between the initial CAM and each pixel's semantic affinities. In addition, a number of studies have introduced a multiscale CAM fusion strategy to complement edge information with high spatial resolution, and these methods have made promising progress [44], [45]. Recently, Fang et al. [46] introduced point labels into CD and proposed a point-based weakly supervised CD method named the consistency-aligned regional growth network. This approach incorporates a point-changed regional growth module designed to generate the expanded annotations from the point annotations.

## III. Methodology

### A. Overview

As depicted in Fig. 1, the proposed method primarily consists of two steps: 1) train classification network and generate pseudo labels and 2) train CD network. Specifically, in the first step, a deep supervision classification network is trained using image-level labels to obtain high-quality multiscale CAM. Subsequently, these multiscale CAM are fused to generate pixel-level pseudo labels. In the second step, the generated pseudolabels are input into the CD network, which is based on a LoRA fine-tuning strategy, to predict the CD results.

### B. Train Classification Network and Generate Pseudolabels

*1) Deep Supervision Classification Network:* Inspired by the prior research on deep supervision [47], which has demonstrated enhanced network performance through the supervision of intermediate layers, we introduce a deep supervision strategy to the weakly supervised CD task. Specifically, we utilize EfficientNetV2-S [48] as the feature extractor to extract multiscale features from bitemporal images. This choice is advantageous due to the model's smaller size and faster training compared with other CNNs [48]. To mitigate the overfitting of the network and conserve computational resources, only the first four convolutional stages of the EfficientNetV2-S are utilized for feature extraction. In addition, we introduce a Siamese architecture to extract independent features from the bitemporal images.

Given bitemporal images with widths $W$ and heights $H$ represented by $I_1, I_2 \in \mathbb{R}^{W\times H\times 3}$, are individually fed into the Siamese network to extract multiscale features, denoted by $f_j^i, i \in 1, 2, 3, 4, j \in 1, 2$, where $i$ denotes the $i$th scale feature and $j$ represents the corresponding image temporal. The channel numbers of the four scales are 24, 24, 48, and 64, respectively, and the resolution sizes are 1/2, 1/2, 1/4, and 1/8 of the input image size, respectively. To obtain discriminative fusion features, we employ our previously proposed feature interleaved fusion module (FIFM) [49], which can enhance the feature fusion by leveraging shared and distinct information, more details can be found in [49]. The fused multiscale features are denoted by $F^i, i \in 1, 2, 3, 4$.

After obtaining the fused features $F^i$ using the FIFM, a classifier unit consisting of a $1 \times 1$ convolutional layer followed by batch normalization (BN) and a rectified linear unit (ReLU) activation is employed to compute the probability scores for the respective classes. The convolutional layer in this study has two output channels, corresponding to the probabilities of the "changed" and "unchanged" categories. These probabilities are then aggregated using GAP, aligning them with the spatial resolution of the image-level labels. The above procedure can be summarized as follows:

$$P = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_1(F))) \tag{1}$$

$$S = \mathrm{Sigmoid}\big(\mathrm{AvgPool}_1(P)\big) \tag{2}$$

where $F$ is one of the fused features generated from FIFM, $\mathrm{Conv}_1$ is a convolutional layer with a size of $1\times1$, $P$ represents the activation map corresponding to the current scale feature, and $S$ denotes the predicted scores. $\mathrm{AvgPool}_1$ is the GAP operation to obtain the probability scores of size $1 \times 1$. Four classifier units with different input channels are utilized for each scale of the fused features.

In this study, we employ the commonly used binary cross-entropy (CE) function as the loss function. In addition, under the deep supervision learning strategy, we compute the sum of the losses for each scale of the fused features to facilitate network convergence. The loss function is formulated as follows:

$$L_{\mathrm{sum}} = -\frac{1}{C}\sum_{k=1}^{K}\sum_{c=1}^{C}\left[T^c\log\left(S^c\right) + \left(1-T^c\right)\log\left(1-S^c\right)\right] \tag{3}$$

where $C$ represents the class number and $C = 2$, $S^c$ is the predicted scores, $T$ denotes the image-level label for the $c$th class, i.e., its value is 1 if there are changes between the bitemporal images; otherwise, its value is 0. $K$ denotes the deep supervision strategy is employed and $K = 4$. $L_{\mathrm{sum}}$ is the total sum of the loss across all feature scales.

*2) Multiscale CAM Fusion:* After the network convergence, we apply the Grad-CAM++ [50] algorithm to generate CAM for each feature scale. Assume that $\tilde{F}^i, 1 \le i \le 4$ denote the output feature maps of the final convolutional layer in each scale, and $\tilde{F}_k^i$ be the $k$th feature channel within $\tilde{F}^i$. For convenience, we use the second-scale feature map $\tilde{F}_k^2$ as an

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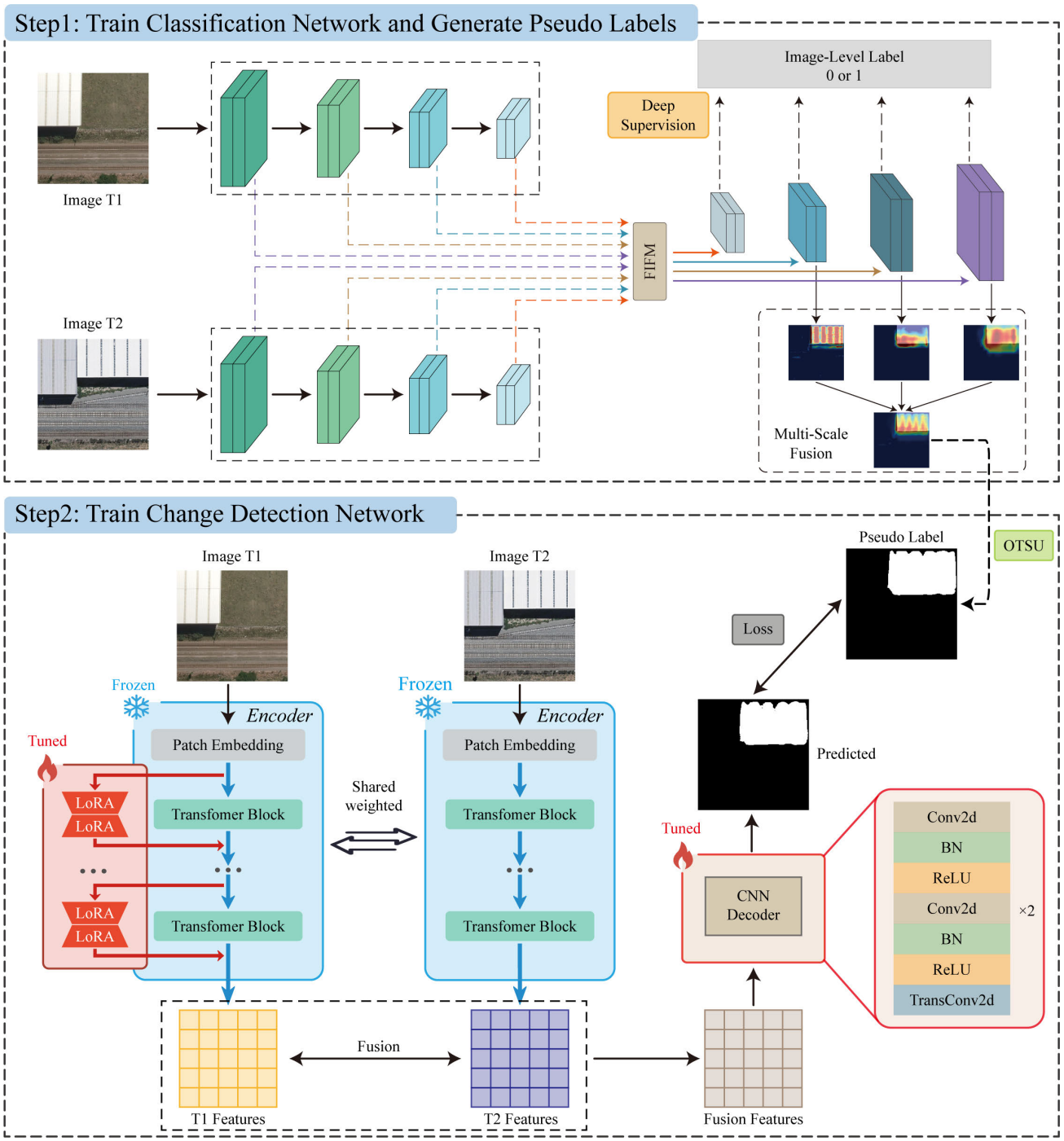



Fig. 1. Framework of the proposed ESAM-CD.

example and further omit the superscript, representing it as $\tilde{F}_k$. The CAM can be computed as follows:

$$\mathrm{CAM}^c = \mathrm{ReLU}\left(\sum_k w_k^c \cdot \tilde{F}_k\right). \tag{4}$$

In the above formula, $c$ represents the changed class, and $\mathrm{CAM}^c$ is the generated CAM for the changed class. $w_k^c$ is the channelwise weight corresponding $\tilde{F}_k$, and it can be computed by

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \cdot \mathrm{ReLU}\left(g_{ij}^{kc}\right) \tag{5}$$

$$g_{ij}^{kc} = \frac{\delta S^c}{\delta \tilde{F}_{ij}^k} \tag{6}$$

$$\alpha_{ij}^{kc} = \frac{\left(g_{ij}^{kc}\right)^2}{2\left(g_{ij}^{kc}\right)^2 + \sum_i \sum_j \tilde{F}_{ij}^k \left(g_{ij}^{kc}\right)^3} \tag{7}$$

where $i$ and $j$ are the spatial location in the feature map $\tilde{F}_k$, and $g_{ij}^{kc}$ is the gradient map of the prediction score $S^c$

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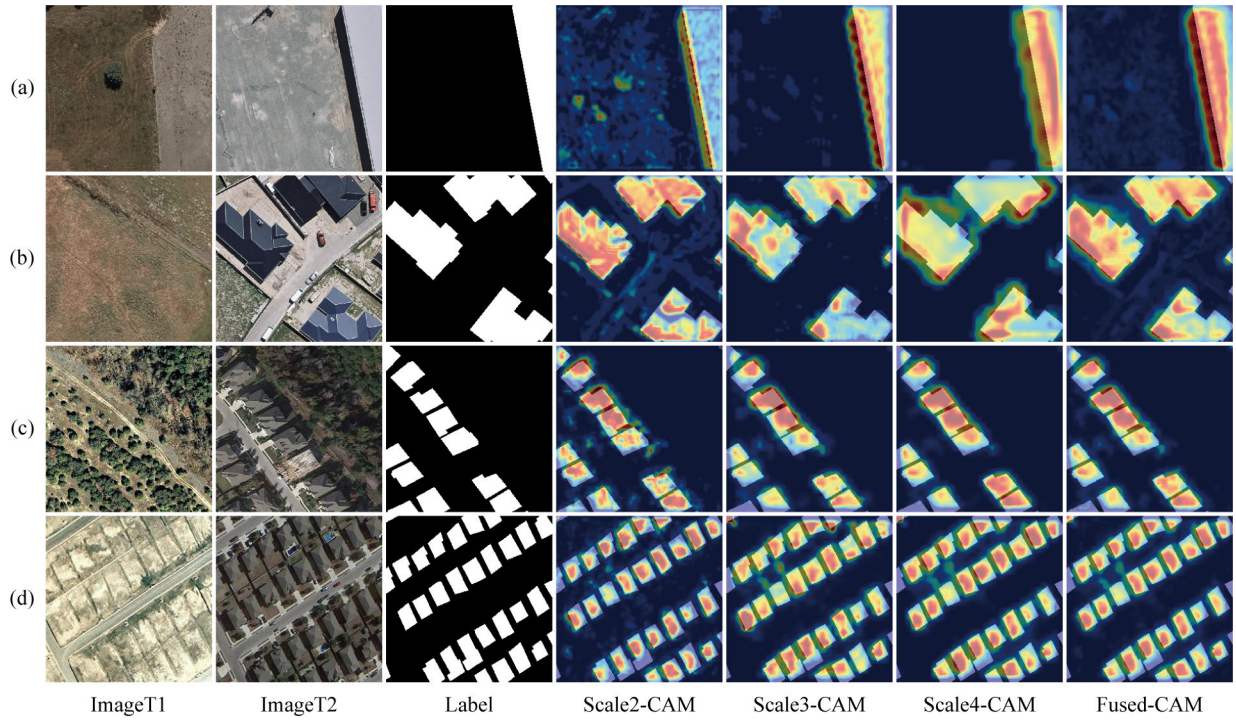



Fig. 2. Visual comparison between single-scale CAM and multiscale fusion CAM. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset.

with respect to the feature map $\tilde{F}_k$. Following this procedure, we can obtain the CAM of the changed class at each scale. However, previous studies [51] have shown that single-scale feature maps may not adequately capture rich spatial details and semantic features. Specifically, low-level feature maps with high resolution contribute to fine-grained details, while high-level feature maps with lower resolution encode semantic features that aid in locating regions of change. Therefore, we introduce a multiscale fusion strategy to integrate the CAM across multiple scales. The fused CAM map can be computed as follows:

$$\mathrm{CAM}_{\mathrm{sum}} = \frac{1}{3}(\mathrm{CAM}_2 + \mathrm{CAM}_3 + \mathrm{CAM}_4) \tag{8}$$

where $\mathrm{CAM}_2$, $\mathrm{CAM}_3$, and $\mathrm{CAM}_4$ represent the generated CAM maps at the 2nd, 3rd, and 4th feature scales, respectively. $\mathrm{CAM}_{\mathrm{sum}}$ is the fused CAM map. The fused multiscale CAM is depicted in Fig. 2. As evident from Fig. 2, while low-scale CAM provides accurate edge details, it also exhibits significant noise responses. Conversely, high-level CAM contains less noise and aids in locating changed regions. The fused multiscale CAM not only preserves accurate edge details but also exhibits diminished noise levels.

It is essential to point out that although the deep supervision strategy is conducted on four features during training, we opt to exclude the CAM map from the first scale in the multiscale CAM fusion process. This decision is based on experimental observations revealing substantial noise in the first scale, which could potentially compromise the accuracy of the fused CAM. Hence, the CAM from the first scale is omitted during multiscale CAM fusion. Fig. 3 presents a visual comparison between CAM from the first and second scales on the two datasets, illustrating the rationale behind excluding CAM from the first scale.

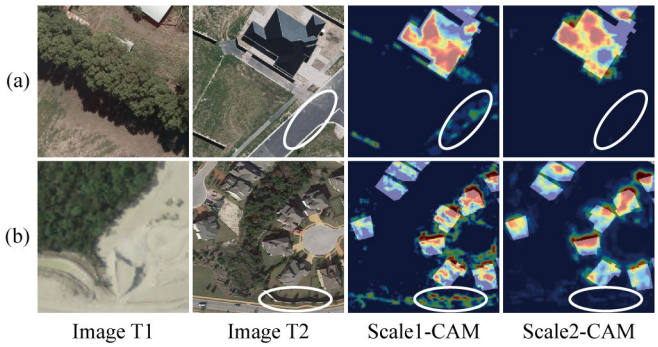



Fig. 3. Visual comparison between Scale1-CAM and Scale2-CAM. (a) WHU-CD dataset. (b) LEVIR-CD dataset.

*3) Pseudolabels Generation:* To derive pixel-level pseudolabels from the fused CAM, a straightforward approach is to empirically select thresholds (such as 0.5) for the CAM. However, applying a fixed threshold across different image pairs can be challenging due to the complexity of the changed objects. Therefore, we introduce the widely employed Otsu segmentation algorithm [52], renowned for its simplicity, robustness, and adaptability. Unlike the use of a fixed threshold, the Otsu algorithm selects the optimal threshold by maximizing the interclass variance of pixel values in the CAM, effectively distinguishing between changed and unchanged regions. Pixels with values higher than the threshold computed by the Otsu are considered changed, while those below it are deemed unchanged. Some examples of the pixel-level pseudolabels generated using the Otsu algorithm are depicted in Fig. 4.

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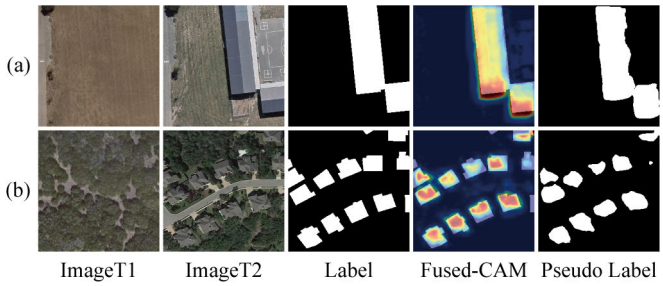



Fig. 4. Comparison between truth and generated pseudolabel using Otsu. (a) WHU-CD dataset. (b) LEVIR-CD dataset.

### C. Train CD Network

*1) Fine-Tune EfficientSAM Based on LoRA:* Although EfficientSAM exhibits excellent performance for natural images, it is still constrained when dealing with complex ground objects in remote sensing images. To improve the feature extraction capability of EfficientSAM on remote sensing images, we introduce the LoRA strategy to fine-tune the image encoder of EfficientSAM. LoRA proposes applying low-rank approximation to delineate parameter updates of weight matrices. Specifically, for a weight matrix $M_0 \in \mathbb{R}^{d\times k}$, LoRA utilizes an additional bypass to update weights via low-rank decomposition, which can be represented as follows:

$$M_0 = \triangle M = M_0 + BA \tag{9}$$

where $B$ and $A$ are the trainable parameters, $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and the rank $r \ll \min(d, k)$. $M_0$ is frozen and does not get gradient updates during training. In practice, we integrate the LoRA technique into each transformer's block of EfficientSAM. The fine-tuning strategy based on LoRA is shown in Fig. 5. Specifically, applying LoRA to the query and value projection layers is a logical choice as it directly affects the attention scores, given that the multihead self-attention mechanism uses cosine similarity to identify the change regions to focus on. The processing can be summarized as follows:

$$\text{Att}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)V \tag{10}$$

$$Q = M_q F_{\text{SAM}} + B_q A_q F_{\text{SAM}} \tag{11}$$

$$K = M_k F_{\text{SAM}} \tag{12}$$

$$V = M_v F_{\text{SAM}} + B_v A_v F_{\text{SAM}} \tag{13}$$

where $M_q$, $M_k$, and $M_v$ represent the frozen weight matrices of projection layers from EfficientSAM. $F_{\text{SAM}}$ is the image feature extracted from EfficientSAM' image encoder. $B_q$, $A_q$, $B_v$, and $A_v$ denote the trainable LoRA parameters. During the training process, the weights of EfficientSAM's image encoder are frozen, while only the parameters of the supplementary LoRA layers are updated. By employing this fine-tuning strategy, we can preserve the powerful image understanding capabilities of EfficientSAM while enhancing its performance on remote sensing images.

To maintain feature independence, we introduce the Siamese architecture to capture the features from bitemporal images. Given a bitemporal image with widths $W$ and heights $H$ represented as $I_1, I_2 \in \mathbb{R}^{R\times H\times 3}$, The bitemporal images are scaled to $1024 \times 1024$ before being individually fed into the frozen Siamese encoder to match the input size of the EfficientSAM encoder. Subsequently, the frozen Siamese encoder with trainable LoRA layers is used to extract features from the bitemporal images. This process can be summarized as follows:

$$\begin{cases} F^1_{\text{SAM}} = \text{SAM}_{\text{LoRA}}(I_1) \\ F^2_{\text{SAM}} = \text{SAM}_{\text{LoRA}}(I_2) \end{cases} \tag{14}$$

where $\text{SAM}_{\text{LoRA}}$ is the frozen EfficientSAM's image encoder with trainable LoRA layers. $F^1_{\text{SAM}}$ and $F^2_{\text{SAM}}$ represent the extracted features from the bitemporal images, respectively, with 256 channels and a scale size of $64 \times 64$. After that, we employ a difference approach to fuse the extracted features.

*2) Prompt-Free Decoder:* While EfficientSAM's image encoder demonstrates excellent feature extraction capabilities, its decoder relies on prompt embedding. To overcome this limitation, we propose a novel prompt-free decoder utilizing the general convolutional layers to generate the CD results. The structure of the proposed decoder based on general convolutional layers is depicted in Fig. 6. Specifically, the proposed decoder consists of two up-sampling blocks with identical structures and a prediction head. Specifically, each up-sampling block primarily consists of two $3 \times 3$ convolutional layers with a stride of 1, followed by a transposed convolutional layer with a stride of 2. The convolutional layers effectively reduce the number of feature channels by half, while the transposed convolutional layer is responsible for up-sampling the spatial resolution of the input features by a factor of 2. Furthermore, the inclusion of BN layers and ReLU activation functions serves to mitigate network overfitting. Employing two blocks of this design enables a gradual reduction in the number of feature channels while simultaneously increasing the spatial resolution. This strategy mitigates information loss caused by rapid reductions in feature channel numbers and swift spatial resolution expansions. Finally, a $3 \times 3$ convolutional layer is applied to generate the change maps. For convenience, the following formulations omit the BN and ReLU layers and can be expressed as follows:

$$\text{Conv}_b = T\text{Conv}(\text{Conv}(\text{Conv}(\bullet))) \tag{15}$$

$$F_{\text{out}} = \text{Conv}_b(\text{Conv}_b(F_{cat})) \tag{16}$$

$$F_{\text{map}} = \text{Conv}_3(F_{\text{out}}) \tag{17}$$

where Conv and $T$Conv are the convolutional and transposed convolutional layers, respectively. $\text{Conv}_b$ is the integrated up-sampling block. $F_{\text{cat}}$ is the concatenated features generated from the Siamese EfficientSAM encoder. $F_{\text{out}}$ is the output feature generated from two up-sampling blocks. $\text{Conv}_3$ represents the final convolutional layer with the output corresponding to the channel number of change classes. $F_{\text{map}}$ denotes the predicted change maps.

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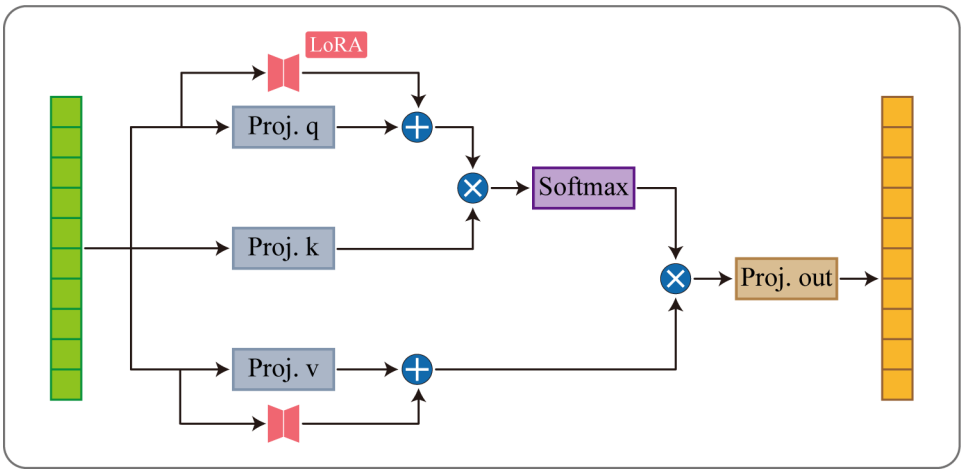



Fig. 5. Structure of the fine-tuning based on LoRA.

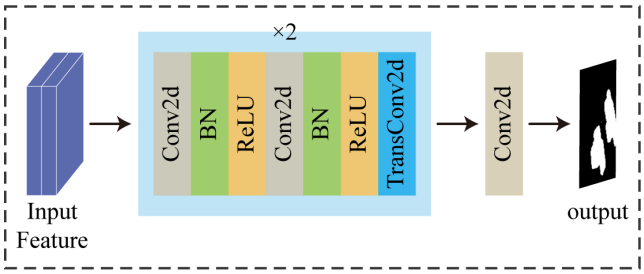



Fig. 6. Structure of the proposed CNN-based decoder.

### D. Loss Function of CD Network

For training the CD network, we employ the CE function as the loss function, which is formulated as follows:

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N} t_i \log(p_i) + (1-t_i)\log(1-p_i) \tag{18}$$

where $N$ is the quantity of pixels that comprise each bitemporal image and $t$ and $p$ are the pseudolabels generated from Section III-B and the predicted maps, respectively. $i$ is the pixel location in the pseudolabels and predicted maps.

## IV. Experiments

This section presents a comprehensive evaluation of the proposed ESAM-CD through a series of experiments. We begin by introducing the datasets used, followed by a comparison with four state-of-the-art (SOTA) methods. Subsequently, we describe the evaluation metrics and implementation details, and finally, we present the experimental results and ablation analysis.

### A. Datasets

The proposed ESAM-CD is validated using two public CD datasets, namely, the WHU-CD [53] and the LEVIR-CD [54] datasets. The details of these datasets and the data split strategy are outlined as follows.

*1) WHU-CD Dataset:* This dataset consists of a pair of satellite images acquired in 2012 and 2016, captured over Christchurch, New Zealand. The bitemporal images consist of 32 507 × 15 354 pixels with a spatial resolution of 0.2 m. They are divided into patches of size 256 × 256 pixels with a 50-pixel overlap. Following the previous study [45], a threshold of 0.14 is set within each patch to distinguish between positive and negative samples. Patches with a changed pixel ratio exceeding 0.14 are labeled as positive (e.g., "1"), while those below are labeled as negative (i.e., "0"). Consequently, a total of 11 618 patches were obtained, consisting of 9896 negative samples and 1722 positive samples. These samples were randomly divided into training, validation, and testing sets using a ratio of 7:1:2. Some samples are shown in Fig. 7.

*2) LEVIR-CD Dataset:* This dataset comprises 637 pairs of Google images, each with a resolution of 0.5 m and a pixel size of 1024 × 1024. The bitemporal images were divided into nonoverlapping patches of size 256 × 256, yielding a total of 10 192 patches. These patches were categorized into 8805 negative samples and 1387 positive samples. Following the split strategy employed for the WHU-CD datasets, a ratio of 7:1:2 was used to randomly divide both positive and negative samples into training, validation, and testing sets. Some samples are shown in Fig. 7.

### B. Benchmarks Methods

To validate the effectiveness of the proposed ESAM-CD, we conducted comparative experiments with several weakly supervised CD methods that offer publicly available code. These methods include W-CDNet [55], fully convolutional change detection (FCD)-GAN [56], TransWCD [57], CS-WSCDNet [19], and MSWSNet [44]. Brief details of these methods are provided as follows:

*1) W-CDNet:* W-CDNet is a weakly supervised method that employs a W-shaped Siamese U-Net architecture for feature extraction from image pairs. The bibranch features are compared using a comparison block to generate a raw change mask. Subsequently, a model called change segmentation and classification (CSC) is introduced to train an accurate mask

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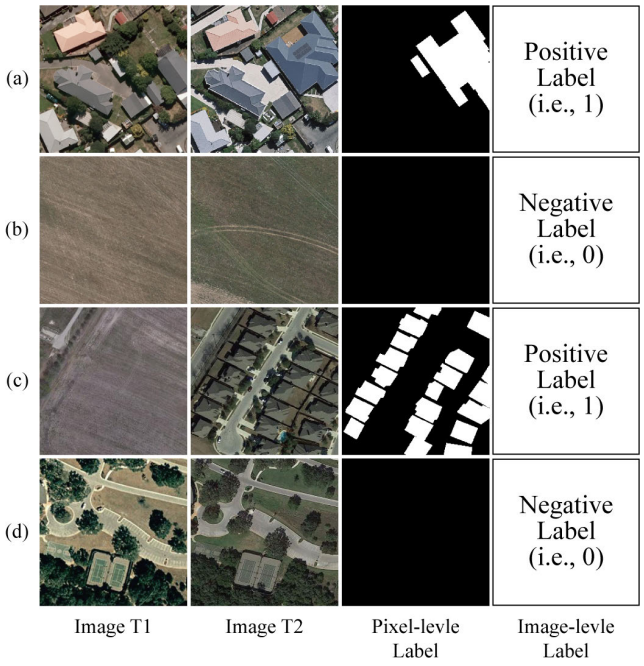



Fig. 7. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset.

and subsequently segment the current input image with this mask.

*2) FCD-GAN:* This method integrates FCD with GAN tailored for weakly supervised CD. Specifically, the FCD-GAN method comprises three core components: a segmentor, a generator, and a discriminator. These elements are integrated within an adversarial learning framework, which is operationalized during the training phase utilizing image-level labels to facilitate the learning process.

*3) TransWCD:* TransWCD is a weakly supervised CD method that employs a hierarchical transformer encoder to extract multiscale features. To incorporate multiscale information, the multiscale CAM scales the last convolutional layers rather than generating them from multistage features. In this article, we use the Siamese dual-stream TransWCD as the comparison method.

*4) CS-WSCDNet:* This method combines the CAM with the SAM for CD. A reactivation branch is introduced to generate recalibrated CAM, followed by utilizing the SAM model as an optimizer to refine the pixel-level pseudolabels.

*5) MSWSNet:* The MSWSNet introduces a multiscale CAM for generating pseudolabels. To address the challenge of inaccurate labels, this method incorporates an adaptive online noise correction module directly into the segmentation step, effectively mitigating the impact of false label generation.

### C. Evaluation Metrics

Following the established evaluation protocol in CD research, we employ a comprehensive set of performance metrics to assess the model's performance, including precision (Pre), recall (Re), $F1$, and intersection over union (IoU). It is essential to emphasize that the $F1$ is selected as the primary evaluation metric, given the inherent constrained relationship between precision and recall. These evaluation metrics are formally defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Re} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F1 = 2 \times \frac{\text{Pre} \times \text{Re}}{\text{Pre} + \text{Re}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{22}$$

where TP represents the correctly predicted changed pixel numbers, FP denotes the unchanged pixels incorrectly predicted as changed, and FN represents the changed pixels missed during detection.

### D. Implementation Details

All experiments were conducted using Pytorch 1.12.0 on an NVIDIA GeForce RTX 3090Ti GPU with 24 GB of memory. As for the pseudolabel generation network, we use AdamW as the optimizer with an initial learning rate of $2 \times 10^{-4}$ and weight decay of $1 \times 10^{-4}$. The batch size and the epochs are set to 8 and 220, respectively. As for CD network training, we froze the EfficientSAM's encoder and fine-tuned the additional LoRA layers and decoder. We use AdamW as the optimizer with an initial learning rate of $1 \times 10^{-4}$ and decay of $1 \times 10^{-3}$. The batch size is set to 4 due to the GPU limits, and the epochs are 50.

### E. Comparison Analysis

*1) Comparison Experiments on CAM Generation:* For weakly supervised CD, the quality of the generated CAM directly affects the corresponding pseudolabels, which in turn impacts the performance of the CD network. Consequently, we first conduct a comparative analysis of the CAM's quality generated by different methods. It is important to point out that W-CDNet and FCD-GAN do not involve CAM generation. Thus, we compare CAM's quality with TransWCD, CS-WSCDNet, and MSWSNet. In addition, we include GradCAM++ [50], a classic CAM generation method, for comparison with the proposed method. It is important to note that, in order to establish a fair comparison, we employ the Otsu algorithm consistently across all comparison methods to evaluate their respective quality and efficacy.

Table I presents the quantitative comparison of the proposed method against other SOTA methods across two datasets. Notably, GradCAM++ exhibits the poorest performance on both datasets, yielding $F1$ scores of 69.72% and 58.27% on WHU-CD and LEVIR-CD, respectively. In contrast, CS-WSCDNet achieves $F1$ scores of 77.32% and 69.10% on the same datasets. These results indicate that CS-WSCDNet significantly outperforms GradCAM++ on both datasets, which may be attributed to the fact that CS-WSCDNet uses the reactivation strategy to enhance the quality of the generated CAM. When comparing two multiscale methods, TransWCD's results are slightly lower than those of MSWSNet. This difference may be attributed to MSWSNet employing multiscale feature

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TABLE I
COMPARISONS OF CAM AMONG DIFFERENT METHODS ON THE TWO DATASETS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

| Methods | WHU-CD | | | | LEVIR-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre (%) | Re (%) | F1 (%) | IoU (%) | Pre (%) | Re (%) | F1 (%) | IoU (%) |
| GradCAM++ | 78.53 | 62.69 | 69.72 | 53.51 | 67.20 | 51.44 | 58.27 | 41.12 |
| TransWCD | 83.25 | 66.45 | 73.91 | 58.62 | 70.34 | 64.73 | 67.42 | 50.86 |
| CS-WSCDNet | 83.08 | 72.31 | 77.32 | 63.02 | 71.93 | 66.48 | 69.10 | 52.79 |
| MSWSNet | 81.37 | 71.52 | 76.13 | 61.46 | 73.76 | 63.56 | 68.28 | 51.83 |
| ESAM-CD | **84.15** | **79.24** | **81.62** | **68.95** | **75.31** | **70.85** | **73.01** | **57.49** |

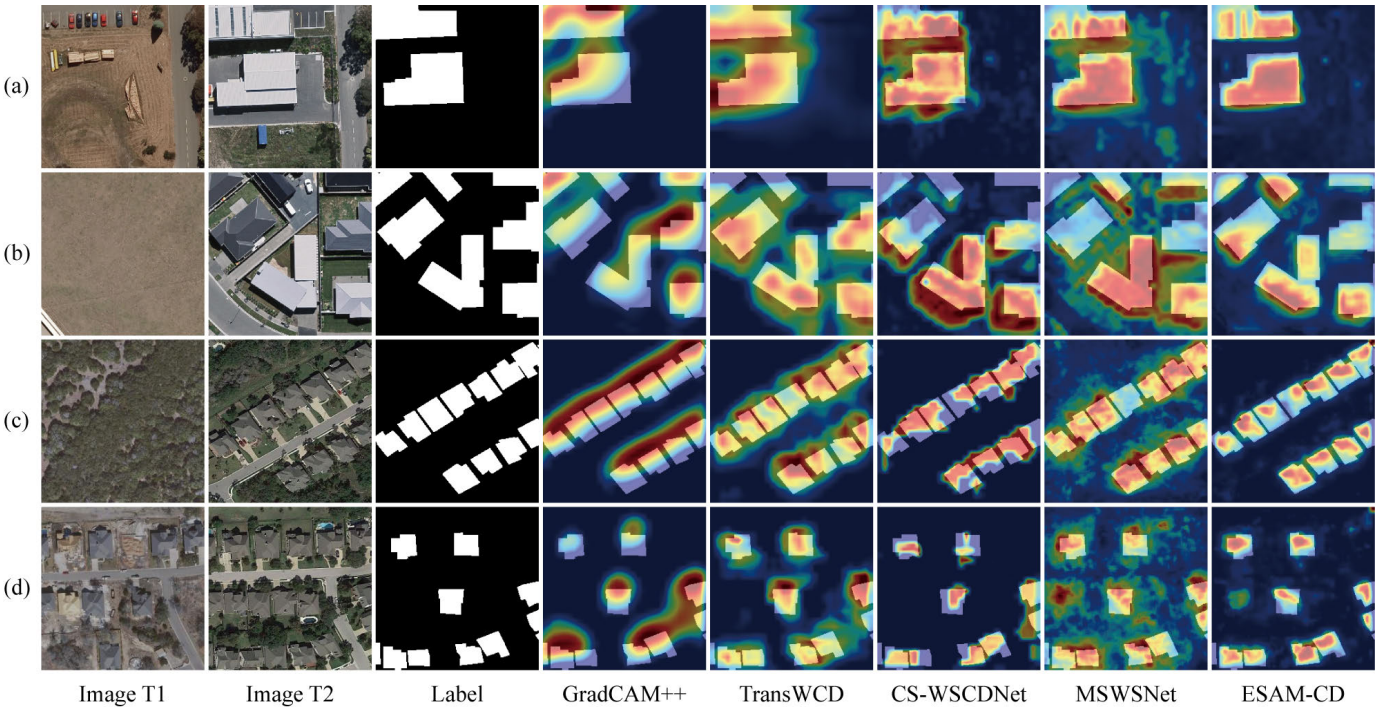



Fig. 8. Visual comparison of CAM among different methods. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset.

layers throughout the network, whereas TransWCD applies multiscale scaling only to the last convolutional layer. Consequently, MSWSNet could capture richer feature information, leading to superior performance. Among all the methods evaluated, the proposed ESAM-CD achieves the best performance on both datasets, with $F1$ scores of 81.62% and 73.01% on WHU-CD and LEVIR-CD, respectively. This superior performance is attributed to ESAM-CD's use of deep supervision and a multiscale fusion strategy, as well as the exclusion of the first-layer CAM, which typically contains excessive noise.

Fig. 8 shows the heatmaps of CAM generated by different methods. It is evident that the CAM generated by GradCAM++ only approximately covers the changed targets. Specifically, the CAM produced by GradCAM++ displays significant overactivation in unchanged areas and underactivation in changed areas. In addition, it struggles to effectively distinguish densely clustered changed regions in the LEVIR-CD dataset. Similarly, the TransWCD results follow trends similar to GradCAM++, possibly because these methods rely solely on the last convolutional layers, which lack detailed information about changed regions. In comparison, MSWSNet more accurately locates the positions of changed targets. However, the CAM generated by MSWSNet contains significant noise. The CAM from CS-WSCDNet shows less overactivation and noise, likely due to its use of deeper features, which result in blurred edges in the changed regions. In contrast, the proposed ESAM-CD method achieves superior visual performance. On the one hand, the activation regions within the CAM of ESAM-CD accurately locate changed regions and preserve more detailed boundary information. On the other hand, the CAM generated by ESAM-CD exhibits less noise than MSWSNet. Overall, this visual comparison aligns with the quantitative results presented in Table I, demonstrating that the proposed ESAM-CD delivers the best performance.

*2) Comparison Experiments on CD:* Table II presents the comprehensive experimental results for each method on the two datasets. It is crucial to emphasize that $F1$ is chosen as the primary evaluation metric for the results. On the WHU-CD dataset, W-CDNet exhibits relatively poor performance, achieving an $F1$ of 52.08% and an IoU of 35.21%. FCD-GAN slightly outperforms W-CDNet, with $F1$ and IoU at 54.36% and 37.33%, respectively. The results of TransWCD, MSWSNet, and CS-WSCDNet perform comparably, with both

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TABLE II
QUANTITATIVE COMPARISON OF EXPERIMENTAL AMONG DIFFERENT METHODS ON THE TWO DATASETS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

| Methods | WHU-CD | | | | LEVIR-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre (%) | Re (%) | F1 (%) | IoU (%) | Pre (%) | Re (%) | F1 (%) | IoU (%) |
| W-CDNet | **73.15** | 40.43 | 52.08 | 35.21 | **69.92** | 27.34 | 39.31 | 24.46 |
| FCD-GAN | 57.84 | 51.28 | 54.36 | 37.33 | 38.47 | 55.73 | 45.52 | 29.47 |
| TransWCD | 67.31 | 62.60 | 64.87 | 48.01 | 52.68 | 56.34 | 54.45 | 37.41 |
| CS-WSCDNet | 70.53 | 66.18 | 68.29 | 51.85 | 60.18 | 55.66 | 57.83 | 40.68 |
| MSWSNet | 63.67 | 68.83 | 66.15 | 49.42 | 62.35 | 48.85 | 54.78 | 37.72 |
| ESAM-CD | 71.52 | **75.36** | **73.39** | **57.97** | 66.13 | **59.18** | **62.46** | **45.41** |

TABLE III
QUANTITATIVE COMPARISON BETWEEN SINGLE LAYER AND MULTISCALE ON THE TWO DATASETS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

| Methods | WHU-CD | | | | LEVIR-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre (%) | Re (%) | F1 (%) | IoU (%) | Pre (%) | Re (%) | F1 (%) | IoU (%) |
| Single-Scale | 69.89 | 75.08 | 72.39 | 56.73 | 71.44 | 52.51 | 60.53 | 43.40 |
| Multi-Scale | **84.15** | **79.24** | **81.62** | **68.95** | **75.31** | **70.85** | **73.01** | **57.49** |

achieving an $F1$ score exceeding 64% and CS-WSCDNet achieving the second-highest score with an $F1$ of 68.29%. Notably, the proposed ESAM-CD outperforms others across recall, $F1$, and IoU metrics, achieving scores of 75.36%, 73.39%, and 57.97%, respectively. In comparison to the second-highest performing method, CS-WSCDNet, the proposed ESAM-CD demonstrates substantial improvements of 5.10% and 6.12% in the $F1$ and IoU metrics, respectively. On the LEVIR-CD dataset, each method exhibits performance trends similar to those observed on the WHU-CD dataset. Notably, the proposed ESAM-CD maintains its status as the best-performing approach, achieving recall, $F1$, and IoU scores of 59.18%, 62.46%, and 45.41%, respectively. CS-WSCDNet ranks as the second-best performer, with an $F1$ score of 57.83% and an IoU of 40.68%. Our method outperforms CS-WSCDNet by a considerable margin, surpassing it by 4.63% in the $F1$ score and 4.73% points in the IoU.

Based on the quantitative results, the inferior performance of W-CDNet may be attributed to the method's reliance solely on posttemporal images in the CSC module to detect changes between bitemporal images. It is important to note that there are various types of changes between bitemporal images, including appearance, disappearances, exchanges, and others [58]. The unsatisfactory results of the FCD-GAN method may stem from the fact that while it leverages the advantage of GAN, it also introduces significant noise, as evident from Figs. 9 and 10. The four methods that utilize CAM outperform both W-CDNet and FCD-GAN. Specifically, the multiscale information in TransWCD is obtained by scaling the last convolutional layer, which results in a loss of rich spatial detail. Consequently, TransWCD produces more false detections and rougher edges. In contrast, MSWSNet utilizes multiscale feature information by incorporating shallow feature layers. As a result, it demonstrates fewer false detections and missed detections compared with TransWCD. However, MSWSNet employs the shallowest features, which contain more noise, thereby affecting the accuracy of the results to some extent. CS-WSCDNet's relatively superior performance may be attributed to its reactivation strategy. In contrast, the proposed ESAM-CD utilizes the multiscale CAM fusion, preserving edge details without introducing significant noise. In addition, the powerful visual recognition and segmentation capabilities of EfficientSAM's image encoder are introduced into CD. In summary, based on the quantitative and qualitative analyses presented above, we conclude that the proposed ESAM-CD outperforms other weakly supervised CD methods in terms of performance.

### F. Ablation Studies

A set of ablation studies is conducted to assess the impact of the multiscale CAM fusion with deep supervision strategy and the trainable LoRA with the prompt-free decoder based on general convolutional layers by systematically removing each component and evaluating the resulting performance.

*1) Ablation Experiment on Multiscale CAM Fusion With Deep Supervision:* Multiscale CAM fusion with deep supervision plays a crucial role in accurate CAM and pseudolabel generation. To better illustrate the effectiveness of multiscale CAM fusion with deep supervision, we compare it with using only the deepest features for CAM and pseudolabel generation. For ease of description, we will refer to the strategy of using only the deepest features as the "single-scale" and multiscale fusion with deep supervision as the "multiscale." Table III presents the quantitative comparison of single-scale and multiscale CAM fusion strategies on both datasets. The results demonstrate the superior performance of the multiscale strategy across all evaluation metrics. Specifically, on the WHU-CD dataset, the multiscale strategy achieves an improvement of 9.23% in $F1$ and a substantial improvement of 12.22% in IoU compared with the single-scale strategy. Similarly, these improvements are consistent on the LEVIR-CD dataset, where the multiscale strategy surpasses the single-scale strategy by 12.48% in $F1$ and 14.09% in IoU, demonstrating its superiority.

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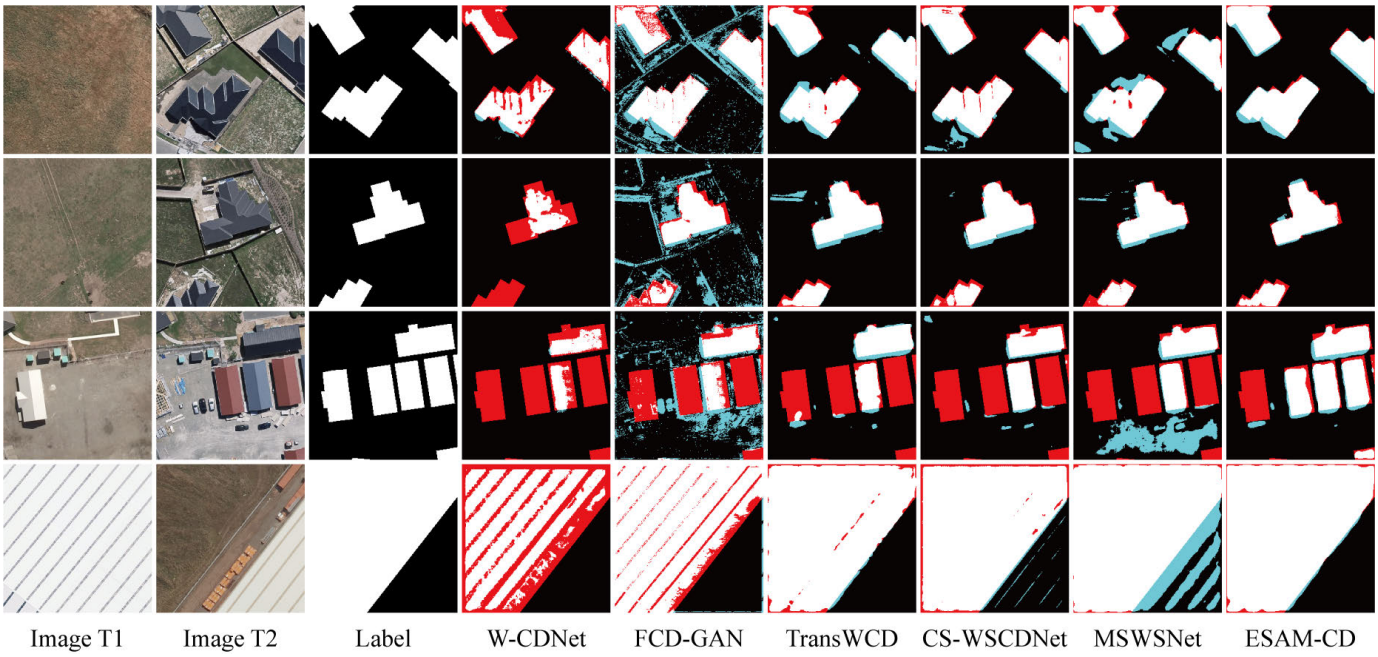



Fig. 9. Comparison of visual results among different methods on the WHU-CD dataset. White represents TP, black represents TN, red represents FN, and blue represents FP.

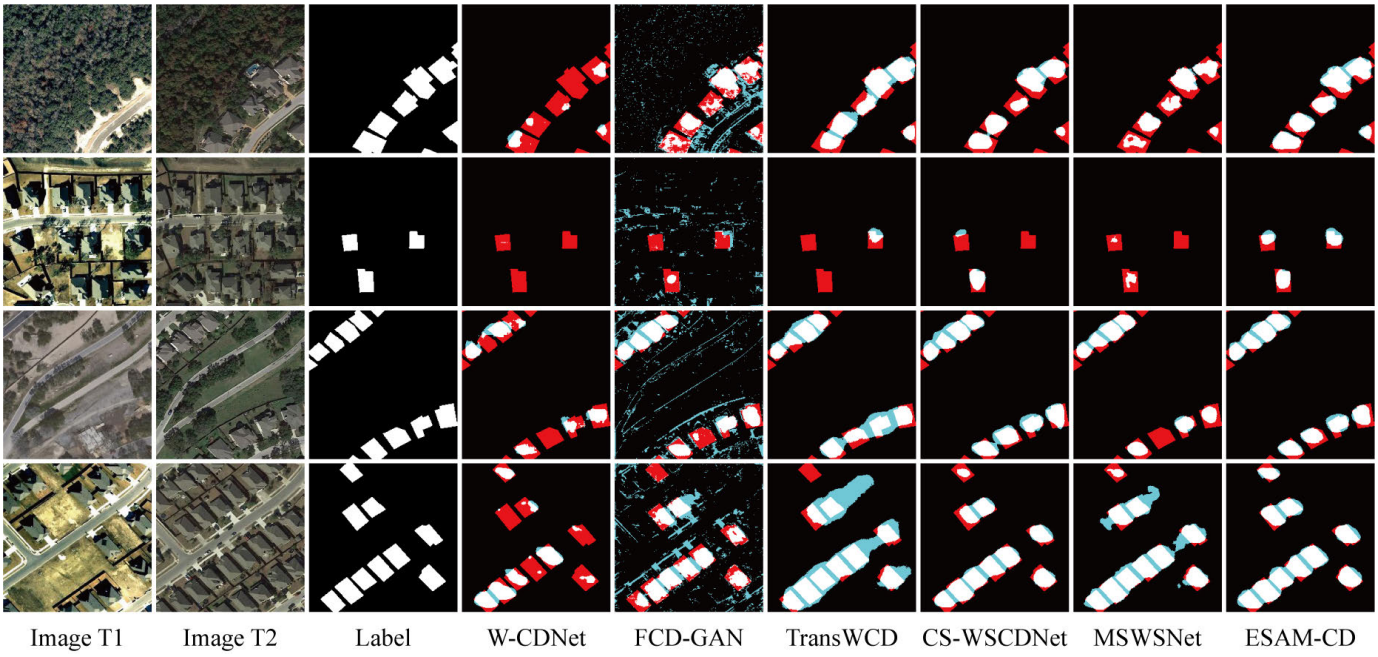



Fig. 10. Comparison of visual results among different methods on the LEVIR-CD dataset. White represents TP, black represents TN, red represents FN, and blue represents FP.

A more intuitive visualization of the comparison between the two methods on the two datasets is presented in Fig. 11. On the WHU-CD dataset, the CAM generated by the single-scale strategy displays numerous over-activation areas and rough edges around changed regions. In contrast, CAM generated by the multiscale fusion strategy with deep supervision accurately pinpoints areas of change. The corresponding generated pseudolabels from the single-scale strategy exhibit significant false and missed detections, whereas those from the multiscale strategy align more closely with the ground-truth labels. On the LEVIR-CD dataset, the CAM from the single-scale strategy shows significant under-activation regions with changes, while those from the multiscale strategy have fewer instances of under-activation, yielding overall satisfactory results. Regarding the generated pseudolabels, the changed regions in the single-scale results appear clustered together, making it difficult to distinguish them effectively, whereas the generated pseudolabels of the multiscale strategy only exhibit slight adhesion.In summary, the comparison illustrates that the proposed multiscale fusion with deep supervision generates higher quality CAM compared with the single-scale

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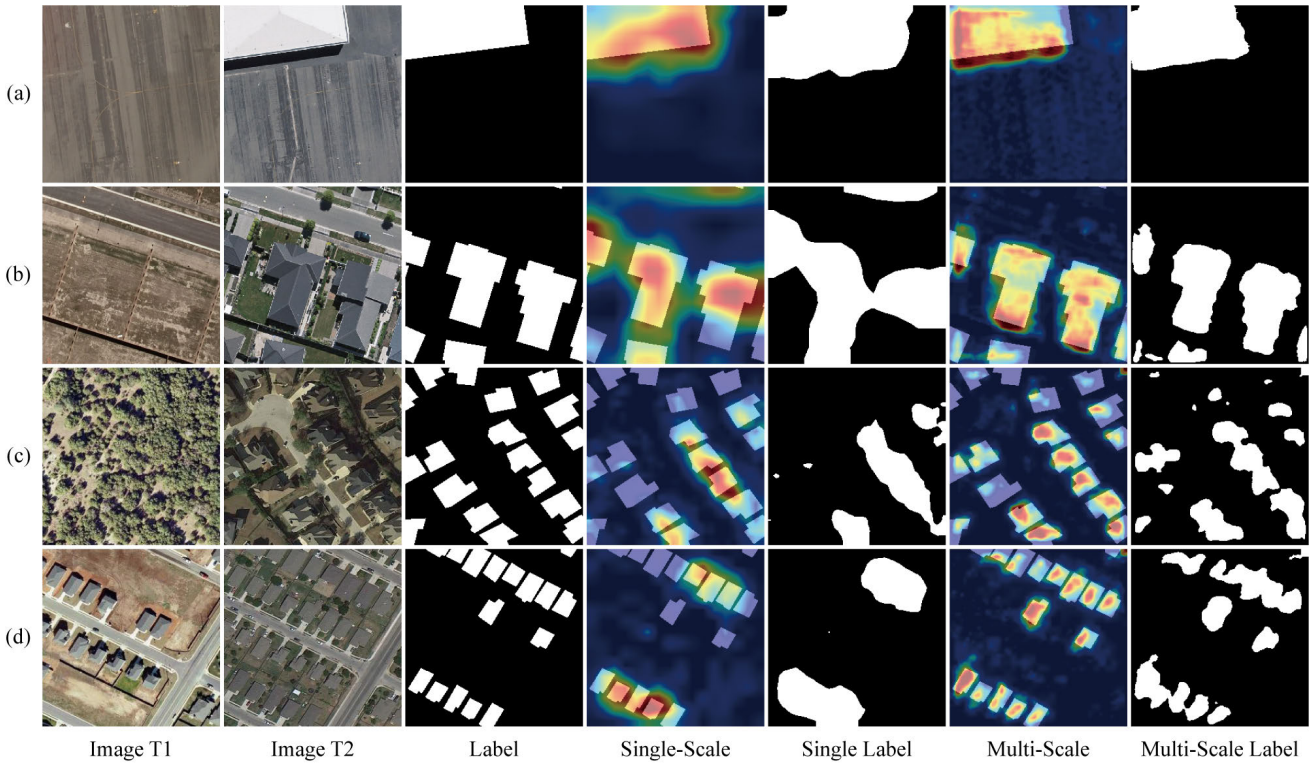



Fig. 11. Visual comparison between single-scale and multiscale methods. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset.

TABLE IV
EXPERIMENTAL RESULTS OF FINE-TUNED EFFICIENTSAM USING LORA ON THE TWO DATASETS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

| Methods | WHU-CD | | | | LEVIR-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre (%) | Re (%) | F1 (%) | IoU (%) | Pre (%) | Re (%) | F1 (%) | IoU (%) |
| LoRA(w/o) | **74.17** | 68.89 | 71.43 | 55.56 | 63.06 | 55.34 | 58.95 | 41.80 |
| LoRA(w) | 71.52 | **75.36** | **73.39** | **57.97** | **66.13** | **59.18** | **62.46** | **45.41** |

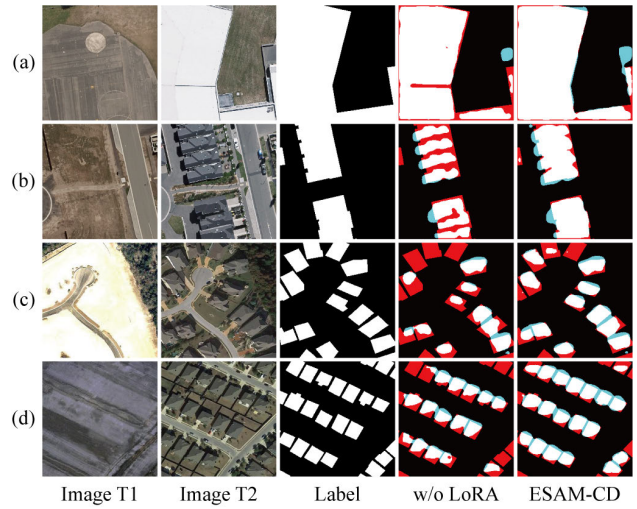



Fig. 12. Visual comparison between single-scale and multiscale methods. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset. White represents TP, black represents TN, red represents FN, and blue represents FP.

strategy, consequently enhancing the accuracy of the generated pseudolabels.

*2) Ablation Experiment on Fine-Tuned EfficientSAM Using LoRA:* To validate the importance of trainable LoRA, an ablation study is conducted by removing LoRA and utilizing the frozen image encoder for feature extraction. Table IV presents the quantitative results of this ablation experiment. The results demonstrate that the use of LoRA consistently yields higher $F1$ and IoU scores on both datasets. Specifically, on the WHU-CD dataset, the method without LoRA obtains the best precision with 74.17%, while the $F1$ and IoU decrease by 1.96% and 2.41%, respectively, compared with the method with LoRA. On the LEVIR-CD dataset, the method with LoRA achieves the best scores across all metrics, with improvements of 3.51% in $F1$ and 3.61% in IoU. The inferior performance of the method without LoRA may be attributed to the fact that the domain information of EfficientSAM's image encoder is primarily based on natural images, which significantly differ from remote sensing images. Therefore, the conclusion can be drawn that the additional trainable LoRA could fine-tune EfficientSAM's image encoder to learn the domain information present in remote sensing images.

As shown in Fig. 12, the visual results of the method without LoRA exhibit more false detections compared with those obtained when using LoRA. Furthermore, it is essential to point out that the results of the method without LoRA show slighter adhesion compared with those obtained when using LoRA. In other words, the method without LoRA

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TABLE V
QUANTITATIVE COMPARISON BETWEEN THE EFFICIENTSAM AND ESAM-CD ON THE TWO DATASETS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

| Methods | WHU-CD | | | | LEVIR-CD | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre (%) | Re (%) | F1 (%) | IoU (%) | Pre (%) | Re (%) | F1 (%) | IoU (%) |
| EfficientSAM | 57.85 | **82.47** | 68.00 | 51.52 | 63.46 | 39.60 | 48.77 | 32.25 |
| ESAM-CD | **71.52** | 75.36 | **73.39** | **57.97** | **66.13** | **59.18** | **62.46** | **45.41** |

excels at splitting each individual target in the remote sensing images, suggesting its proficiency in splitting each individual target in the remote sensing images. This phenomenon may be attributed to the freezing of the encoder during training. In summary, the method with LoRA demonstrates superior performance, underscoring the significance of trainable LoRA as a key component in the network.

*3) Ablation Experiment for Prompt-Free Decoder:* An important contribution of this study is the integration of EfficientSAM's image encoder with general convolutional layers for CD tasks. To evaluate the effectiveness of the proposed prompt-free decoder, we compare its performance against the EfficientSAM's default structure. Specifically, the bitemporal images are input into the frozen EfficientSAM's image encoder, which uses a Siamese architecture to extract features. The bitemporal features are then fused using a difference approach. Finally, we train the EfficientSAM decoder to produce the CD results.

The comparison results between EfficientSAM and the proposed method are presented in Table V. It is evident that the proposed prompt-free decoder achieves higher performance in terms of $F1$ and IoU on both datasets. Specifically, compared with directly employing EfficientSAM's decoder, there is an increase of at least 5.39% in $F1$ score and 6.45% in IoU on the WHU-CD dataset. Regarding the LEVIR-CD dataset, the proposed method achieves significant improvements of approximately 13.69% and 13.16% in $F1$ score and IoU, respectively. Fig. 13 provides a visual comparison between EfficientSAM and the proposed method. The EfficientSAM results show different performance levels for WHU-CD and LEVIR-CD. Specifically, there are more false detections in the results of EfficientSAM for WHU-CD, while more missed detections occur in the results for LEVIR-CD. This discrepancy may be due to the difficulty of directly applying EfficientSAM's segment decoder to CD tasks. In contrast, the proposed method demonstrates superior performance.

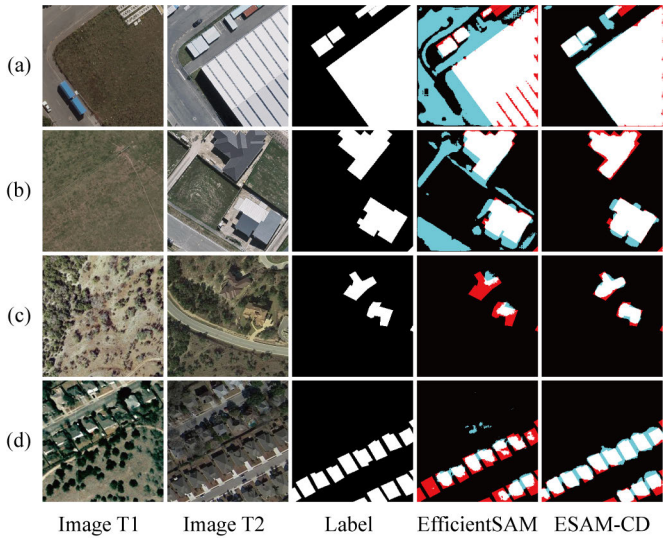



Fig. 13. Visual comparison between single-scale and multiscale methods. (a) and (b) Samples of the WHU-CD dataset. (c) and (d) Samples of the LEVIR-CD dataset. White represents TP, black represents TN, red represents FN, and blue represents FP.

## V. CONCLUSION

This article proposes ESAM-CD, a novel weakly supervised CD method that is based on multiscale CAM fusion and a fine-tuned EfficientSAM. To overcome the need for extensive pixel-level labels, which are both time-consuming and labor-intensive, a multiscale CAM fusion strategy integrated with deep supervision is proposed to obtain CAM and further generate accurate pixel-level pseudo labels from image-level labels. Furthermore, to explore the powerful zero-shot feature extraction capability of EfficientSAM's image encoder for remote sensing image CD, we fine-tuned it using LoRA. In addition, a prompt-free decoder based on general convolutional layers is proposed to alleviate the reliance of EfficientSAM's decoder on prompts. Experimental results on two widely used CD datasets, WHU-CD and LEVIR-CD, demonstrate that ESAM-CD outperforms SOTA weakly supervised methods by 5.10% and 4.63% in terms of $F1$, respectively. Moreover, ablation experiments validate the efficacy of multiscale CAM fusion based on a deep supervision strategy and underscore the potential of integrating the fine-tuned EfficientSAM using LoRA and general convolutional layers decoders in CD tasks. Therefore, we hope our work will inspire further research in the field of CD.

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